# Frontend Product Requirements Document (PRD)

**Product:** VidAgent AI **Target:** A hyper-fast, menu-less, agent-driven video editor.

## 1. The Brutal Truth & Core Philosophy

Traditional NLEs (Non-Linear Editors) are dead for 90% of creators. Our UI must reflect this.

- **No complex toolbars.** If a user has to search for a "razor tool," we failed.
- **Total Transparency.** The AI must never be a "black box." The user must see exactly what the AI is doing in real-time.
- **Speed over everything.** The UI must feel as fast as texting.

## 2. Global Design System (The Vibe)

The app must look like a professional, futuristic workspace. Dark mode is mandatory (editors hate bright screens).

- **Background (The Void):** Obsidian Black ( `#0B0C10` ) - Deep, distraction-free.
- **Surface Panels:** Dark Grey ( `#1F2833` ) - For chat boxes and timeline tracks.
- **AI Active Accent:** Electric Cyan ( `#66FCF1` ) - Used *only* when the AI is actively processing, thinking, or highlighting a change it made.
- **Text/Icons:** Almost White ( `#C5C6C7` ) for primary text, Slate ( `#8A9CA8` ) for secondary text.
- **Typography:** * *UI Elements:* Inter (Clean, modern, highly legible).
  - *Agent Output/Logs:* JetBrains Mono (Gives a technical, "under-the-hood" feel to AI actions).

## 3. Global Layout Architecture

A clean, 2-pane split screen on desktop.

- **Left Pane (35% width): The Command Center.** This is where the user chats with VidAgent.
- **Right Pane (65% width): The Glass Box.** This holds the Video Preview Player (top) and the Live Timeline (bottom).
- **Mobile Experience:** The Chat interface is the *only* thing visible. The video preview sits at the top, and the timeline is hidden behind an "Advanced Edit" toggle.

## 4. Core Component Specifications

### A. The Command Center (Chat Interface)

This replaces the traditional menu bar.

- **The Input Box:** A sticky, auto-expanding text box at the bottom. Must include a highly visible Microphone icon for voice prompts.
- **The Action Feed (Chat History):** * Do *not* use generic loading spinners. Use skeleton text and "typing" effects.
  - When the agent executes a tool, show a beautiful success checklist:
    - `<Icon> Removed 47 silences (saved 3m 12s)`
    - `<Icon> Color graded (Cinematic preset)`
- **Clarification Cards:** If the AI needs input (e.g., "Which language for captions?"), it renders a UI card with quick-select buttons directly in the chat, not a popup modal.

**B. The Video Player (Preview)**

- **Minimalist Controls:** Play, Pause, Fullscreen, and a custom aspect-ratio toggle (16:9, 9:16, 1:1).
- **Overlay Bounding Boxes:** When the AI does "Smart Reframing" or "Object Removal", show a faint, dashed Electric Cyan box tracking the subject on the video canvas.

**C. The "Glass Box" Timeline This is for verification, not manual scrubbing (though manual scrubbing is allowed).**

- **Track Layout:** Stacked horizontally (Video, Audio, Music, Captions).
- **AI Edit Highlights:** Every time the AI makes a cut or removes a filler word, that specific splice on the timeline must glow slightly with the Electric Cyan accent color.
- **Hover-to-Explain: (CRITICAL)** When the user hovers over an AI cut, a sleek tooltip must instantly appear: *"Removed 1.5s silence"* or *"Ducked background music by -12db"*.
- **One-Click Reversals:** Clicking any AI-edited clip brings up a micro-menu: `[Undo this cut] | [Adjust manually]` .

**D. The Drop Zone (Onboarding)**

- When the app opens, the Right Pane is completely empty except for a massive, dashed drop zone.
- Text: "Drag raw footage here, or paste a YouTube URL."

## 5. Micro-Interactions & Animation

- **Speed:** All UI transitions (opening panels, hovering buttons) must be snappy (100ms - 150ms). No slow, floaty animations.
- **Agent 'Thinking' State:** When Gemini is processing a complex prompt, the border of the Right Pane (The Glass Box) should pulse softly with the Cyan accent to show the entire app is working on the video.
- **Drag and Drop:** Users should be able to drag an image from their desktop directly into the chat and say, "Make the color grade look like this."

## 6. Recommended Frontend Tech Stack

To execute this fast and keep it scalable:

- **Framework:** Next.js (React) - For fast routing and component architecture.
- **Styling:** Tailwind CSS - For rapid UI iteration and strict design system adherence.
- **State Management:** Zustand - Lightweight and perfect for managing complex timeline states without Redux boilerplate.
- **Video/Timeline Rendering:** Wavesurfer.js (for drawing fast audio waveforms) combined with HTML5 `<canvas>` or a customized Video.js player for the timeline visualizer.
- **Animations:** Framer Motion - For the smooth chat list updates and timeline highlight pulses.